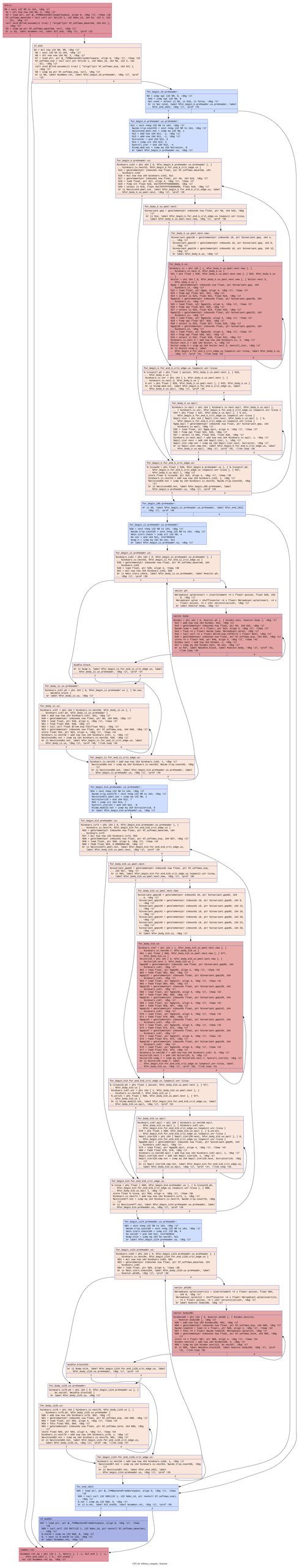

```
entry:
%0 = sext i32 %M to i64, !dbg !17
 %1 = shl nuw nsw i64 %0, 2, !dbg !17
 %2 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !17, !tbaa !18
 %T_softmax_maxelem = tail call ptr %2(i32 1, i32 %dev_id, i64 %1, i32 2, i32
... 32), !dbg !17
 call void @llvm.assume(i1 true) [ "align"(ptr %T_softmax_maxelem, i64 64) ],
... !dbg !17
 %3 = icmp eq ptr %T_softmax_maxelem, null, !dbg !17
 br i1 %3, label %common.ret, label %if_end, !dbg !17, !prof !22

if_end:
%4 = mul nsw i32 %N, %M, !dbg !17
 %5 = sext i32 %4 to i64, !dbg !17
 %6 = shl nuw nsw i64 %5, 2, !dbg !17
 %7 = load ptr, ptr @__TVMBackendAllocWorkspace, align 8, !dbg !17, !tbaa !18
 %T_softmax_exp = tail call ptr %7(i32 1, i32 %dev_id, i64 %6, i32 2, i32
... 32), !dbg !17
 call void @llvm.assume(i1 true) [ "align"(ptr %T_softmax_exp, i64 64) ],
... !dbg !17
 %8 = icmp eq ptr %T_softmax_exp, null, !dbg !17
 br i1 %8, label %common.ret, label %for_begin_i0.preheader, !dbg !17, !prof
... !22

for_begin_i0.preheader:
%9 = icmp sgt i32 %M, 0, !dbg !17
 %10 = icmp sgt i32 %N, 0
 %or.cond = select i1 %9, i1 %10, i1 false, !dbg !17
 br i1 %or.cond, label %for_begin_k.preheader.us.preheader, label
... %for_end_i022, !dbg !17, !prof !25

for_begin_k.preheader.us.preheader:
%11 = zext nneg i32 %N to i64, !dbg !17
 %wide.trip.count55 = zext nneg i32 %M to i64, !dbg !17
 %exitcond.peel.not = icmp eq i32 %N, 1
 %12 = add nsw i64 %11, -1, !dbg !17
 %13 = add nsw i64 %11, -2, !dbg !17
 %xtraiter = and i64 %12, 3
 %14 = icmp ult i64 %13, 3
 %unroll_iter = and i64 %12, -4
 %lcmp.mod.not = icmp eq i64 %xtraiter, 0
 br label %for_begin_k.preheader.us, !dbg !17

for_begin_k.preheader.us:
%indvars.iv52 = phi i64 [ 0, %for_begin_k.preheader.us.preheader ], [
... %indvars.iv.next53, %for_begin_k.for_end_k_crit_edge.us ]
 %15 = getelementptr inbounds nuw float, ptr %T_softmax_maxelem, i64
... %indvars.iv52
 %16 = mul nuw nsw i64 %indvars.iv52, %11
 %17 = getelementptr inbounds nuw float, ptr %A, i64 %16, !dbg !17
 %18 = load float, ptr %17, align 4, !dbg !17, !tbaa !27
 %19 = fcmp olt float %18, 0xC7EFFFFFE0000000, !dbg !17
 %20 = select i1 %19, float 0xC7EFFFFFE0000000, float %18, !dbg !17
 br i1 %exitcond.peel.not, label %for_begin_k.for_end_k_crit_edge.us, label
... %for_body_k.us.peel.next, !dbg !17, !prof !29

for_body_k.us.peel.next:
%invariant.gep = getelementptr inbounds nuw float, ptr %A, i64 %16, !dbg
... !17
 br i1 %14, label %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa,
... label %for_body_k.us.peel.next.new, !dbg !17, !prof !30

for_body_k.us.peel.next.new:
%invariant.gep130 = getelementptr inbounds i8, ptr %invariant.gep, i64 4,
... !dbg !17
 %invariant.gep132 = getelementptr inbounds i8, ptr %invariant.gep, i64 8,
... !dbg !17
 %invariant.gep134 = getelementptr inbounds i8, ptr %invariant.gep, i64 12,
... !dbg !17
 br label %for_body_k.us, !dbg !17

for_body_k.us:
%indvars.iv = phi i64 [ 1, %for_body_k.us.peel.next.new ], [
... %indvars.iv.next.3, %for_body_k.us ]
 %21 = phi float [ %20, %for_body_k.us.peel.next.new ], [ %33, %for_body_k.us
... ]
 %niter = phi i64 [ 0, %for_body_k.us.peel.next.new ], [ %niter.next.3,
... %for_body_k.us ]
 %gep = getelementptr inbounds nuw float, ptr %invariant.gep, i64
... %indvars.iv, !dbg !17
 %22 = load float, ptr %gep, align 4, !dbg !17, !tbaa !27
 %23 = fcmp ogt float %21, %22, !dbg !17
 %24 = select i1 %23, float %21, float %22, !dbg !17
 %gep131 = getelementptr inbounds float, ptr %invariant.gep130, i64
... %indvars.iv, !dbg !17
 %25 = load float, ptr %gep131, align 4, !dbg !17, !tbaa !27
 %26 = fcmp ogt float %24, %25, !dbg !17
 %27 = select i1 %26, float %24, float %25, !dbg !17
 %gep133 = getelementptr inbounds float, ptr %invariant.gep132, i64
... %indvars.iv, !dbg !17
 %28 = load float, ptr %gep133, align 4, !dbg !17, !tbaa !27
 %29 = fcmp ogt float %27, %28, !dbg !17
 %30 = select i1 %29, float %27, float %28, !dbg !17
 %gep135 = getelementptr inbounds float, ptr %invariant.gep134, i64
... %indvars.iv, !dbg !17
 %31 = load float, ptr %gep135, align 4, !dbg !17, !tbaa !27
 %32 = fcmp ogt float %30, %31, !dbg !17
 %33 = select i1 %32, float %30, float %31, !dbg !17
 %indvars.iv.next.3 = add nuw nsw i64 %indvars.iv, 4, !dbg !17
 %niter.next.3 = add i64 %niter, 4, !dbg !17
 %niter.ncmp.3 = icmp eq i64 %niter.next.3, %unroll_iter, !dbg !17
 br i1 %niter.ncmp.3, label
... %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa, label %for_body_k.us,
... !dbg !17, !prof !31, !llvm.loop !32

for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa:
%.lcssa117.ph = phi float [ poison, %for_body_k.us.peel.next ], [ %33,
... %for_body_k.us ]
 %indvars.iv.unr = phi i64 [ 1, %for_body_k.us.peel.next ], [
... %indvars.iv.next.3, %for_body_k.us ]
 %.unr = phi float [ %20, %for_body_k.us.peel.next ], [ %33, %for_body_k.us ]
 br i1 %lcmp.mod.not, label %for_begin_k.for_end_k_crit_edge.us, label
... %for_body_k.us.epil, !dbg !17, !prof !34

for_body_k.us.epil:
%indvars.iv.epil = phi i64 [ %indvars.iv.next.epil, %for_body_k.us.epil ],
... [ %indvars.iv.unr, %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa ]
 %34 = phi float [ %37, %for_body_k.us.epil ], [ %.unr,
... %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa ]
 %epil.iter = phi i64 [ %epil.iter.next, %for_body_k.us.epil ], [ 0,
... %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa ]
 %gep.epil = getelementptr inbounds nuw float, ptr %invariant.gep, i64
... %indvars.iv.epil, !dbg !17
 %35 = load float, ptr %gep.epil, align 4, !dbg !17, !tbaa !27
 %36 = fcmp ogt float %34, %35, !dbg !17
 %37 = select i1 %36, float %34, float %35, !dbg !17
 %indvars.iv.next.epil = add nuw nsw i64 %indvars.iv.epil, 1, !dbg !17
 %epil.iter.next = add i64 %epil.iter, 1, !dbg !17
 %epil.iter.cmp.not = icmp eq i64 %epil.iter.next, %xtraiter, !dbg !17
 br i1 %epil.iter.cmp.not, label %for_begin_k.for_end_k_crit_edge.us, label
... %for_body_k.us.epil, !dbg !17, !prof !35, !llvm.loop !36

for_begin_k.for_end_k_crit_edge.us:
%.lcssa49 = phi float [ %20, %for_begin_k.preheader.us ], [ %.lcssa117.ph,
... %for_begin_k.for_end_k_crit_edge.us.loopexit.unr-lcssa ], [ %37,
... %for_body_k.us.epil ], !dbg !17
 store float %.lcssa49, ptr %15, align 4, !dbg !17, !tbaa !38
 %indvars.iv.next53 = add nuw nsw i64 %indvars.iv52, 1, !dbg !17
 %exitcond56.not = icmp eq i64 %indvars.iv.next53, %wide.trip.count55, !dbg
... !17
 br i1 %exitcond56.not, label %for_begin_i06.preheader, label
... %for_begin_k.preheader.us, !dbg !17, !prof !29

for_begin_i06.preheader:
br i1 %9, label %for_begin_i1.preheader.us.preheader, label %for_end_i022,
... !dbg !17, !prof !40

for_begin_i1.preheader.us.preheader:
%38 = zext nneg i32 %N to i64, !dbg !17
 %wide.trip.count65 = zext nneg i32 %M to i64, !dbg !17
 %min.iters.check = icmp ult i32 %N, 4
 %n.vec = and i64 %11, 2147483644
 %cmp.n = icmp eq i64 %n.vec, %11
 br label %for_begin_i1.preheader.us, !dbg !17

for_begin_i1.preheader.us:
%indvars.iv62 = phi i64 [ 0, %for_begin_i1.preheader.us.preheader ], [
... %indvars.iv.next63, %for_begin_i1.for_end_i1_crit_edge.us ]
 %39 = getelementptr inbounds nuw float, ptr %T_softmax_maxelem, i64
... %indvars.iv62
 %40 = load float, ptr %39, align 4, !tbaa !38
 %41 = mul nuw nsw i64 %indvars.iv62, %38
 br i1 %min.iters.check, label %for_body_i1.us.preheader, label %vector.ph,
... !dbg !17, !prof !30

vector.ph:
%broadcast.splatinsert = insertelement <4 x float> poison, float %40, i64
... 0, !dbg !17
 %broadcast.splat = shufflevector <4 x float> %broadcast.splatinsert, <4 x
... float> poison, <4 x i32> zeroinitializer, !dbg !17
 br label %vector.body, !dbg !17

vector.body:
%index = phi i64 [ 0, %vector.ph ], [ %index.next, %vector.body ], !dbg !17
 %42 = add nuw nsw i64 %index, %41, !dbg !17
 %43 = getelementptr inbounds nuw float, ptr %A, i64 %42, !dbg !17
 %wide.load = load <4 x float>, ptr %43, align 4, !dbg !17, !tbaa !27
 %44 = fsub <4 x float> %wide.load, %broadcast.splat, !dbg !17
 %45 = tail call <4 x float> @llvm.exp.v4f32(<4 x float> %44), !dbg !17
 %46 = getelementptr inbounds nuw float, ptr %T_softmax_exp, i64 %42, !dbg !17
 store <4 x float> %45, ptr %46, align 4, !dbg !17, !tbaa !42
 %index.next = add nuw i64 %index, 4, !dbg !17
 %47 = icmp eq i64 %index.next, %n.vec, !dbg !17
 br i1 %47, label %middle.block, label %vector.body, !dbg !17, !prof !31,
... !llvm.loop !44

middle.block:
br i1 %cmp.n, label %for_begin_i1.for_end_i1_crit_edge.us, label
... %for_body_i1.us.preheader, !dbg !17, !prof !47

for_body_i1.us.preheader:
%indvars.iv57.ph = phi i64 [ 0, %for_begin_i1.preheader.us ], [ %n.vec,
... %middle.block ]
 br label %for_body_i1.us, !dbg !17

for_body_i1.us:
%indvars.iv57 = phi i64 [ %indvars.iv.next58, %for_body_i1.us ], [
... %indvars.iv57.ph, %for_body_i1.us.preheader ]
 %48 = add nuw nsw i64 %indvars.iv57, %41, !dbg !17
 %49 = getelementptr inbounds nuw float, ptr %A, i64 %48, !dbg !17
 %50 = load float, ptr %49, align 4, !dbg !17, !tbaa !27
 %51 = fsub float %50, %40, !dbg !17
 %52 = tail call float @llvm.exp.f32(float %51), !dbg !17
 %53 = getelementptr inbounds nuw float, ptr %T_softmax_exp, i64 %48, !dbg !17
 store float %52, ptr %53, align 4, !dbg !17, !tbaa !42
 %indvars.iv.next58 = add nuw nsw i64 %indvars.iv57, 1, !dbg !17
 %exitcond61.not = icmp eq i64 %indvars.iv.next58, %38, !dbg !17
 br i1 %exitcond61.not, label %for_begin_i1.for_end_i1_crit_edge.us, label
... %for_body_i1.us, !dbg !17, !prof !48, !llvm.loop !49

for_begin_i1.for_end_i1_crit_edge.us:
%indvars.iv.next63 = add nuw nsw i64 %indvars.iv62, 1, !dbg !17
 %exitcond66.not = icmp eq i64 %indvars.iv.next63, %wide.trip.count65, !dbg
... !17
 br i1 %exitcond66.not, label %for_begin_k14.preheader.us.preheader, label
... %for_begin_i1.preheader.us, !dbg !17, !prof !29

for_begin_k14.preheader.us.preheader:
%54 = zext nneg i32 %N to i64, !dbg !17
 %wide.trip.count76 = zext nneg i32 %M to i64, !dbg !17
 %exitcond71.peel.not = icmp eq i32 %N, 1
 %xtraiter119 = and i64 %12, 7
 %55 = icmp ult i64 %13, 7
 %unroll_iter124 = and i64 %12, -8
 %lcmp.mod122.not = icmp eq i64 %xtraiter119, 0
 br label %for_begin_k14.preheader.us, !dbg !17

for_begin_k14.preheader.us:
%indvars.iv73 = phi i64 [ 0, %for_begin_k14.preheader.us.preheader ], [
... %indvars.iv.next74, %for_begin_k14.for_end_k16_crit_edge.us ]
 %56 = getelementptr inbounds nuw float, ptr %T_softmax_maxelem, i64
... %indvars.iv73
 %57 = mul nuw nsw i64 %indvars.iv73, %54
 %58 = getelementptr inbounds nuw float, ptr %T_softmax_exp, i64 %57, !dbg !17
 %59 = load float, ptr %58, align 4, !dbg !17, !tbaa !42
 %60 = fadd float %59, 0.000000e+00, !dbg !17
 br i1 %exitcond71.peel.not, label %for_begin_k14.for_end_k16_crit_edge.us,
... label %for_body_k15.us.peel.next, !dbg !17, !prof !29

for_body_k15.us.peel.next:
%invariant.gep95 = getelementptr inbounds nuw float, ptr %T_softmax_exp,
... i64 %57, !dbg !17
 br i1 %55, label %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa,
... label %for_body_k15.us.peel.next.new, !dbg !17, !prof !30

for_body_k15.us.peel.next.new:
%invariant.gep136 = getelementptr inbounds i8, ptr %invariant.gep95, i64
... 4, !dbg !17
 %invariant.gep138 = getelementptr inbounds i8, ptr %invariant.gep95, i64 8,
... !dbg !17
 %invariant.gep140 = getelementptr inbounds i8, ptr %invariant.gep95, i64 12,
... !dbg !17
 %invariant.gep142 = getelementptr inbounds i8, ptr %invariant.gep95, i64 16,
... !dbg !17
 %invariant.gep144 = getelementptr inbounds i8, ptr %invariant.gep95, i64 20,
... !dbg !17
 %invariant.gep146 = getelementptr inbounds i8, ptr %invariant.gep95, i64 24,
... !dbg !17
 %invariant.gep148 = getelementptr inbounds i8, ptr %invariant.gep95, i64 28,
... !dbg !17
 br label %for_body_k15.us, !dbg !17

for_body_k15.us:
%indvars.iv67 = phi i64 [ 1, %for_body_k15.us.peel.next.new ], [
... %indvars.iv.next68.7, %for_body_k15.us ]
 %61 = phi float [ %60, %for_body_k15.us.peel.next.new ], [ %77,
... %for_body_k15.us ]
 %niter125 = phi i64 [ 0, %for_body_k15.us.peel.next.new ], [
... %niter125.next.7, %for_body_k15.us ]
 %gep96 = getelementptr inbounds nuw float, ptr %invariant.gep95, i64
... %indvars.iv67, !dbg !17
 %62 = load float, ptr %gep96, align 4, !dbg !17, !tbaa !42
 %63 = fadd float %62, %61, !dbg !17
 %gep137 = getelementptr inbounds float, ptr %invariant.gep136, i64
... %indvars.iv67, !dbg !17
 %64 = load float, ptr %gep137, align 4, !dbg !17, !tbaa !42
 %65 = fadd float %64, %63, !dbg !17
 %gep139 = getelementptr inbounds float, ptr %invariant.gep138, i64
... %indvars.iv67, !dbg !17
 %66 = load float, ptr %gep139, align 4, !dbg !17, !tbaa !42
 %67 = fadd float %66, %65, !dbg !17
 %gep141 = getelementptr inbounds float, ptr %invariant.gep140, i64
... %indvars.iv67, !dbg !17
 %68 = load float, ptr %gep141, align 4, !dbg !17, !tbaa !42
 %69 = fadd float %68, %67, !dbg !17
 %gep143 = getelementptr inbounds float, ptr %invariant.gep142, i64
... %indvars.iv67, !dbg !17
 %70 = load float, ptr %gep143, align 4, !dbg !17, !tbaa !42
 %71 = fadd float %70, %69, !dbg !17
 %gep145 = getelementptr inbounds float, ptr %invariant.gep144, i64
... %indvars.iv67, !dbg !17
 %72 = load float, ptr %gep145, align 4, !dbg !17, !tbaa !42
 %73 = fadd float %72, %71, !dbg !17
 %gep147 = getelementptr inbounds float, ptr %invariant.gep146, i64
... %indvars.iv67, !dbg !17
 %74 = load float, ptr %gep147, align 4, !dbg !17, !tbaa !42
 %75 = fadd float %74, %73, !dbg !17
 %gep149 = getelementptr inbounds float, ptr %invariant.gep148, i64
... %indvars.iv67, !dbg !17
 %76 = load float, ptr %gep149, align 4, !dbg !17, !tbaa !42
 %77 = fadd float %76, %75, !dbg !17
 %indvars.iv.next68.7 = add nuw nsw i64 %indvars.iv67, 8, !dbg !17
 %niter125.next.7 = add i64 %niter125, 8, !dbg !17
 %niter125.ncmp.7 = icmp eq i64 %niter125.next.7, %unroll_iter124, !dbg !17
 br i1 %niter125.ncmp.7, label
... %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa, label
... %for_body_k15.us, !dbg !17, !prof !50, !llvm.loop !51

for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa:
%.lcssa116.ph = phi float [ poison, %for_body_k15.us.peel.next ], [ %77,
... %for_body_k15.us ]
 %indvars.iv67.unr = phi i64 [ 1, %for_body_k15.us.peel.next ], [
... %indvars.iv.next68.7, %for_body_k15.us ]
 %.unr121 = phi float [ %60, %for_body_k15.us.peel.next ], [ %77,
... %for_body_k15.us ]
 br i1 %lcmp.mod122.not, label %for_begin_k14.for_end_k16_crit_edge.us, label
... %for_body_k15.us.epil, !dbg !17, !prof !52

for_body_k15.us.epil:
%indvars.iv67.epil = phi i64 [ %indvars.iv.next68.epil,
... %for_body_k15.us.epil ], [ %indvars.iv67.unr,
... %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa ]
 %78 = phi float [ %80, %for_body_k15.us.epil ], [ %.unr121,
... %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa ]
 %epil.iter120 = phi i64 [ %epil.iter120.next, %for_body_k15.us.epil ], [ 0,
... %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa ]
 %gep96.epil = getelementptr inbounds nuw float, ptr %invariant.gep95, i64
... %indvars.iv67.epil, !dbg !17
 %79 = load float, ptr %gep96.epil, align 4, !dbg !17, !tbaa !42
 %80 = fadd float %79, %78, !dbg !17
 %indvars.iv.next68.epil = add nuw nsw i64 %indvars.iv67.epil, 1, !dbg !17
 %epil.iter120.next = add i64 %epil.iter120, 1, !dbg !17
 %epil.iter120.cmp.not = icmp eq i64 %epil.iter120.next, %xtraiter119, !dbg
... !17
 br i1 %epil.iter120.cmp.not, label %for_begin_k14.for_end_k16_crit_edge.us,
... label %for_body_k15.us.epil, !dbg !17, !prof !47, !llvm.loop !53

for_begin_k14.for_end_k16_crit_edge.us:
%.lcssa = phi float [ %60, %for_begin_k14.preheader.us ], [ %.lcssa116.ph,
... %for_begin_k14.for_end_k16_crit_edge.us.loopexit.unr-lcssa ], [ %80,
... %for_body_k15.us.epil ], !dbg !17
 store float %.lcssa, ptr %56, align 4, !dbg !17, !tbaa !38
 %indvars.iv.next74 = add nuw nsw i64 %indvars.iv73, 1, !dbg !17
 %exitcond77.not = icmp eq i64 %indvars.iv.next74, %wide.trip.count76, !dbg
... !17
 br i1 %exitcond77.not, label %for_begin_i124.preheader.us.preheader, label
... %for_begin_k14.preheader.us, !dbg !17, !prof !29

for_begin_i124.preheader.us.preheader:
%81 = zext nneg i32 %N to i64, !dbg !17
 %wide.trip.count86 = zext nneg i32 %M to i64, !dbg !17
 %min.iters.check104 = icmp ult i32 %N, 4
 %n.vec107 = and i64 %11, 2147483644
 %cmp.n114 = icmp eq i64 %n.vec107, %11
 br label %for_begin_i124.preheader.us, !dbg !17

for_begin_i124.preheader.us:
%indvars.iv83 = phi i64 [ 0, %for_begin_i124.preheader.us.preheader ], [
... %indvars.iv.next84, %for_begin_i124.for_end_i126_crit_edge.us ]
 %82 = mul nuw nsw i64 %indvars.iv83, %81
 %83 = getelementptr inbounds nuw float, ptr %T_softmax_maxelem, i64
... %indvars.iv83
 %84 = load float, ptr %83, align 4, !tbaa !38
 br i1 %min.iters.check104, label %for_body_i125.us.preheader, label
... %vector.ph105, !dbg !17, !prof !30

vector.ph105:
%broadcast.splatinsert111 = insertelement <4 x float> poison, float %84,
... i64 0, !dbg !17
 %broadcast.splat112 = shufflevector <4 x float> %broadcast.splatinsert111,
... <4 x float> poison, <4 x i32> zeroinitializer, !dbg !17
 br label %vector.body108, !dbg !17

vector.body108:
%index109 = phi i64 [ 0, %vector.ph105 ], [ %index.next113,
... %vector.body108 ], !dbg !17
 %85 = add nuw nsw i64 %index109, %82, !dbg !17
 %86 = getelementptr inbounds nuw float, ptr %T_softmax_exp, i64 %85, !dbg !17
 %wide.load110 = load <4 x float>, ptr %86, align 4, !dbg !17, !tbaa !42
 %87 = fdiv <4 x float> %wide.load110, %broadcast.splat112, !dbg !17
 %88 = getelementptr inbounds nuw float, ptr %T_softmax_norm, i64 %85, !dbg
... !17
 store <4 x float> %87, ptr %88, align 4, !dbg !17, !tbaa !54
 %index.next113 = add nuw i64 %index109, 4, !dbg !17
 %89 = icmp eq i64 %index.next113, %n.vec107, !dbg !17
 br i1 %89, label %middle.block102, label %vector.body108, !dbg !17, !prof
... !31, !llvm.loop !56

middle.block102:
br i1 %cmp.n114, label %for_begin_i124.for_end_i126_crit_edge.us, label
... %for_body_i125.us.preheader, !dbg !17, !prof !47

for_body_i125.us.preheader:
%indvars.iv78.ph = phi i64 [ 0, %for_begin_i124.preheader.us ], [
... %n.vec107, %middle.block102 ]
 br label %for_body_i125.us, !dbg !17

for_body_i125.us:
%indvars.iv78 = phi i64 [ %indvars.iv.next79, %for_body_i125.us ], [
... %indvars.iv78.ph, %for_body_i125.us.preheader ]
 %90 = add nuw nsw i64 %indvars.iv78, %82, !dbg !17
 %91 = getelementptr inbounds nuw float, ptr %T_softmax_exp, i64 %90, !dbg !17
 %92 = load float, ptr %91, align 4, !dbg !17, !tbaa !42
 %93 = fdiv float %92, %84, !dbg !17
 %94 = getelementptr inbounds nuw float, ptr %T_softmax_norm, i64 %90, !dbg
... !17
 store float %93, ptr %94, align 4, !dbg !17, !tbaa !54
 %indvars.iv.next79 = add nuw nsw i64 %indvars.iv78, 1, !dbg !17
 %exitcond82.not = icmp eq i64 %indvars.iv.next79, %81, !dbg !17
 br i1 %exitcond82.not, label %for_begin_i124.for_end_i126_crit_edge.us,
... label %for_body_i125.us, !dbg !17, !prof !48, !llvm.loop !57

for_begin_i124.for_end_i126_crit_edge.us:
%indvars.iv.next84 = add nuw nsw i64 %indvars.iv83, 1, !dbg !17
 %exitcond87.not = icmp eq i64 %indvars.iv.next84, %wide.trip.count86, !dbg
... !17
 br i1 %exitcond87.not, label %for_end_i022, label
... %for_begin_i124.preheader.us, !dbg !17, !prof !29

for_end_i022:
%95 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !17, !tbaa
... !18
 %96 = tail call i32 %95(i32 1, i32 %dev_id, ptr nonnull %T_softmax_exp),
... !dbg !17
 %.not = icmp eq i32 %96, 0, !dbg !17
 br i1 %.not, label %if_end29, label %common.ret, !dbg !17, !prof !58

if_end29:
%97 = load ptr, ptr @__TVMBackendFreeWorkspace, align 8, !dbg !17, !tbaa
... !18
 %98 = tail call i32 %97(i32 1, i32 %dev_id, ptr nonnull %T_softmax_maxelem),
... !dbg !17
 %.not34 = icmp ne i32 %98, 0, !dbg !17
 %. = sext i1 %.not34 to i32, !dbg !17
 br label %common.ret, !dbg !17

common.ret:
%common.ret.op = phi i32 [ -1, %entry ], [ -1, %if_end ], [ -1,
... %for_end_i022 ], [ %., %if_end29 ]
 ret i32 %common.ret.op, !dbg !17
```

CFG for 'softmax_compute_' function